

1. What is the result of the following operation in Python: 17/2 ?

1 point

- ☒ 8 or 8.5, depending on the Python version
- ☐ 8.5
- ☐ 8.0
- ☐ 8

2. Given the following code in Python:

1 point

```
1 >>> mydna = 'acgt'  
2 >>> mydna = mydna + mydna
```

What will be the result of typing the following at the Python interpreter prompt:

>>> myDna

- ☐ 'ACGTACGT'
- ☐ 'acgtacgt'
- ☒ an error message
- ☐ no output

3. The following commands are entered at the prompt of Python interpreter.

1 point

```
1 >>> dna="atgctggggact"  
2 >>> dna[:3]  
3 >>> dna
```

What will be the output of the last command?

- ☐ 'atg'
- ☒ 'atgctggggact'
- ☐ 'ctggggact'
- ☐ 'atgc'

4. What is the output of 'dna'+1+2+3 ?

1 point

- ☐ dna
- ☒ Error
- ☐ dna123
- ☐ dna6

5. Given a string variable called dna, for instance:

1 point

```
>>> dna='agcagtttagcta'
```

What is a correct way to count the number of occurrences of 'ag' in dna:

- ☐ dna.count('a')+dna.count('A')+dna.count('g')+dna.count('G')
- ☐ dna.count('ag')+dna.count('AG')
- ☒ dna.count('ag')
- ☐ dna.count('a')+dna.count('g')

6. What is the value of the variable seqlen, after the following code is entered in Python:

1 point

```
1 >>> seqlen = '10bp'
2 >>> seqlen='2'+seqlen
3 >>> seqlen=seqlen*2
```

- ☒ '210bp210bp'
- ☐ '1010bp'
- ☐ '10bp'
- ☐ '24bp'

7. You wish to display the following text using the print function in Python:

1 point

```
2 GGCAGATCCCCCTAGACODE
```

Select the correct way to display this output in Python 3.xx:

- ☐ print('>HSBGPG Human bone gla gene\transcript "BGP"
GGCAGATCCCCCTAGA')
- ☒ print('>HSBGPG Human bone gla gene\\transcript "BGP"\nGGCAGATCCCCCTAGA')
- ☐ print("""\n>HSBGPG Human bone gla gene\transcript "BGP"
GGCAGATCCCCCTAGA
""")
- ☐ print(">HSBGPG Human bone gla gene\transcript "BGP"\nGGCAGATCCCCCTAGA")

8. A student is writing Python 3.xx code to read in a dna sequence using the following command:

1 point

```
>>> dna=input("Enter a DNA sequence, please:")
```

The student tries three different ways to compute the index of the second occurrence of the string 'atg' in the dna sequence:

```
1 A.  
2  
3 >>> o1 = dna.find('atg')  
4 >>> dna.find('atg',o1+1)  
5  
6 B.  
7  
8 >>> dna.rfind('atg')  
9  
10 C.  
11 >>> dna.find('atg',dna.find('atg')+1)
```

Which of these ways is correct:

- ☒ A or C
☐ A
☐ B
☐ A or B or C

9. What are the types of the following literals, in order?

1 point

1, 1., 1.0, 1e10, 0x12, "1", "1.0", 10000000000000000000, 10000000000000000000.0

- ☒ int, float, float, float, int, str, str, int, float
☐ int, no type (error), float, double, int, string, string, long,
double
☐ int, float, float, float, int, int, float, int, float
☐ int, no type (error), float, float, hex, string, string, int, float

10. What is the result of `int(4+6/2+2*2)`?

1 point

- ☐ 9.0
☐ 9
☒ 11
☐ 11.0